

Data Analysis Report

Project Overview:

This project aims to analyze an e-commerce sales dataset containing 34500 transactions between 2023 and 2025. The goal of the analysis is to understand customer behavior, identify key business insights, detect issues, and provide recommendations to improve sales performance and profitability.

Problem statement:

Despite generating approximately 5.87M in total sales, the business faces lower profitability due to high discount usage, increasing return rates, and uneven regional performance.

- Higher discounts reducing profit margins
- A 6% return rate causing revenue loss
- Weak performance in the Central region
- High return rates in Fashion and Electronics categories

Dataset Selection:

E-commerce Sales Transactions Dataset, which included 34500 records from different product categories and regions, was selected because it contains information related to sales, customers, discounts, profits, and products.

The dataset was gathered from Kaggle

<https://www.kaggle.com/datasets/miadul/e-commerce-sales-transactions-dataset/data>

Data Description:

The dataset contains 17 attributes

Attribute	Description	Data Type
Order id	Unique order identifier	object
Customer id	Unique customer identifier	object
Product id	Unique product identifier	object
category	Product category	object
price	Product price	Numerical(float)
discount	Applied discount	Numerical(float)
quantity	Quantity purchased	Numerical (Integer)
Payment method	Payment type	categorical
Delivery times days	Delivery duration	Numerical (Integer)
Region	Customer region	object
Returned	Return status	object
Total amount	Final order amount	Numerical(float)
Shipping cost	Shipping expenses	Numerical(float)
Profit margin	Profit generated	Numerical(float)
Customer age	Customer age	Numerical(integer)
Order date	Transaction date	Date/Time
Customer gender	Customer gender	object

- No missing values
- There is no inconsistency data

Preparing the Data and cleaning:

- Loading dataset.
- Show its information and statistics.
- Check missing values.
- Check duplicate values.
- Convert date to datetime
- Check inconsistent values.
- Detected outliers using the IQR method.

The code:

```
import numpy as np
import pandas as pd
```

```
df=pd.read_csv("/kaggle/input/datasets/ayagamal11/e-commerce-dataset/ecommerce_sales_34500.csv")
```

```
df.head()
```

	order_id	customer_id	product_id	category	price	discount	quantity	payment_method	order_date	delivery_time_days	region
0	O100000	C17270	P234890	Home	164.08	0.15	1	Credit Card	2023-12-23	4	West
1	O100001	C17603	P228204	Grocery	24.73	0.00	1	Credit Card	2025-04-03	6	South
2	O100002	C10860	P213892	Electronics	175.58	0.05	1	Credit Card	2024-10-08	4	North
3	O100003	C15390	P208689	Electronics	63.67	0.00	1	UPI	2024-09-14	6	South
4	O100004	C15226	P228063	Home	16.33	0.15	1	COD	2024-12-21	6	East

In [105]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 34500 entries, 0 to 34499
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   order_id              34500 non-null  object 
1   customer_id           34500 non-null  object 
2   product_id            34500 non-null  object 
3   category              34500 non-null  object 
4   price                 34500 non-null  float64
5   discount              34500 non-null  float64
6   quantity              34500 non-null  int64  
7   payment_method        34500 non-null  object 
8   order_date            34500 non-null  object 
9   delivery_time_days    34500 non-null  int64  
10  region                34500 non-null  object 
11  returned              34500 non-null  object 
12  total_amount          34500 non-null  float64
13  shipping_cost         34500 non-null  float64
14  profit_margin         34500 non-null  float64
15  customer_age          34500 non-null  int64  
16  customer_gender       34500 non-null  object 
dtypes: float64(5), int64(3), object(9)
memory usage: 4.5+ MB
```

In [106]:

```
df.describe()
```

Out[106]:

	price	discount	quantity	delivery_time_days	total_amount	shipping_cost	profit_margin	customer_age
count	34500.000000	34500.000000	34500.000000	34500.000000	34500.000000	34500.000000	34500.000000	34500.000000
mean	110.391832	0.049291	1.490725	4.814203	170.008494	6.152120	28.118505	43.474377
std	195.620477	0.089884	0.932270	1.242141	357.503014	2.389539	53.352947	14.980882
min	1.010000	0.000000	1.000000	3.000000	0.820000	0.000000	-6.200000	18.000000
25%	16.890000	0.000000	1.000000	4.000000	19.710000	4.420000	1.500000	31.000000
50%	45.890000	0.000000	1.000000	5.000000	56.820000	6.090000	10.550000	43.000000
75%	130.950000	0.100000	2.000000	6.000000	166.530000	7.830000	33.132500	56.000000
max	2930.470000	0.300000	5.000000	13.000000	12931.800000	15.850000	1536.170000	69.000000

In [107]:

```
df.dtypes
```

Out[107]:

```
order_id          object
customer_id       object
product_id        object
category          object
price            float64
discount          float64
quantity          int64
payment_method    object
order_date        object
delivery_time_days int64
region            object
returned          object
total_amount      float64
shipping_cost     float64
profit_margin     float64
customer_age      int64
customer_gender   object
dtype: object
```

```
In [108]: df.isnull()
```

Out[108]:

	order_id	customer_id	product_id	category	price	discount	quantity	payment_method	order_date	delivery_time_days	region	ret
0	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...
34495	False	False	False	False	False	False	False	False	False	False	False	False
34496	False	False	False	False	False	False	False	False	False	False	False	False
34497	False	False	False	False	False	False	False	False	False	False	False	False
34498	False	False	False	False	False	False	False	False	False	False	False	False
34499	False	False	False	False	False	False	False	False	False	False	False	False

34500 rows x 17 columns

```
In [109]: print(df.isnull().sum())
```

```
order_id          0
customer_id       0
product_id        0
category          0
price             0
discount          0
quantity          0
payment_method    0
order_date        0
delivery_time_days 0
region            0
returned          0
total_amount      0
shipping_cost     0
profit_margin     0
customer_age      0
customer_gender   0
dtype: int64
```

```
In [110]: df.duplicated().sum()
```

Out[110]: np.int64(0)

```
In [111]: df['order_date'] = pd.to_datetime(df['order_date'])
print(df['order_date'])
```

```
0      2023-12-23
1      2025-04-03
2      2024-10-08
3      2024-09-14
4      2024-12-21
...
34495  2025-08-13
34496  2023-12-04
34497  2025-02-01
34498  2024-03-27
34499  2025-09-11
Name: order_date, Length: 34500, dtype: datetime64[ns]
```

Defining KPIs:

- Total sales.
- Total profit.
- Total orders.
- Average discount.
- Return Rate.
- Revenue lost to returned orders.

Hypothesis-Driven Analysis:

1. Higher discounts lead to lower profits.
2. Orders without discounts generate higher profitability.
3. Longer delivery times increase return rates.
4. Distant regions have lower sales performance and higher return rates due to longer delivery times and shipping challenges.

Exploratory Data Analysis (EDA):

It is performed using pandas, Excel pivot tables, and power BI to determine the patterns, anomalies, trends, and relationships.

1. Filtering data based on:
 - Profit (`df[df['profit_margin'] > 150]`),
(`df[df['profit_margin'] < 0]`)
 - Discounts (`df[df['discount'] > 0.15]`)
 - Delivery times days (`df[df['delivery_time_days'] > 10]`)
2. Grouping the data based on:
 - sales by region
(`df.groupby('region')['total_amount'].sum()`)

- profit margins by category
(df.groupby('category')['profit_margin'].mean())
- orders by payment method
(df.groupby('payment_method')['order_id'].count())
- Group returns by category
(df.groupby('category')['returned'].value_counts())
- sales by customer gender
(df.groupby('customer_gender')['total_amount'].sum())

3. Aggregation based on:

- Sum (df['total_amount'].sum())
- Mean (df['profit_margin'].mean())
- Count (df['order_id'].count())
- Value counts (df['category'].value_counts()
(df['region'].value_counts(),
df['customer_gender'].value_counts()))
- Multiple agg(sum-mean,count)
(df.groupby('category').agg({'total_amount':
'sum','profit_margin': 'mean','order_id': 'count'}))

4. Comparisons Between Categories:

- Compare sales across categories
(df.groupby('category')['total_amount'].sum())
- Compare return rates across categories
(df.groupby('category')['returned_flag'].mean())
- Compare average profit margins
(df.groupby('category')['profit_margin'].mean())

5. Trend Analysis:

- Monthly orders trend
`df['order_month'] = df['order_date'].dt.month`
`print(df['order_month'])`
- Yearly orders trend
`df['order_year'] = df['order_date'].dt.year`
`print(df['order_year'])` Orders trend per year

patterns

- **Electronics generated the highest sales revenue.**
- **Orders with no discounts produced higher profit margins.**
- **Male customers purchased more Electronics and Sports products.**
- **Longer delivery times were associated with higher return rates.**

Anomalies

- Orders with delivery times greater than 10 days
- Negative profit orders
- High-value outlier orders in Electronics category

Relationships Between Variables

- Higher discounts lead to lower profit margins
- Longer delivery times cause higher return rates
- Regional differences affected sales performance

Identifying Root Causes

1. **Low Profitability in Grocery Category:** it has the lowest profit margins compared to other categories, and many grocery orders generate very little profit or even losses.

This happened because:

- Grocery products have low prices, so the profit earned per order is already small.
- Shipping costs take a large portion of the product price.
- Adding discounts on low-cost products reduces profit even more.
- Combining shipping costs and discounts sometimes causes negative profit margins.

2. High Return Rates in Fashion and Electronics: they have the highest return rates on the platform.

This happened because:

- Customers may return Fashion products because of sizing or fitting issues.
- Electronics products may not meet customer expectations or may have technical problems.
- High discounts can encourage impulsive purchases, increasing the chance of returns.
- Longer delivery times may also increase customer dissatisfaction and returns.

3. Weak Performance in Central Region: it has lower sales and profitability compared to other regions.

This happened because:

- The region records fewer orders than South and North regions.
- Delivery performance may be weaker in this region.
- Customer demand appears lower compared to other regions.

- Limited marketing or operational efficiency may affect regional performance.

Insights

- Male customers dominate in Electronics (38%) and Sports (17%), while female customers lead in Fashion (31%) and Beauty (28%).
- Electronics and Home categories have the highest average profit margins across all regions, while Grocery consistently have the lowest profitability.
- East and South regions perform better across most categories, while the Central region shows weaker profitability and sales performance.
- Fashion and Electronics had the highest return rates, which negatively affected total profit.

Recommendations

- Reduce high discount campaigns and focus on controlled discounts to improve profit margins in Grocery products to reduce losses.
- Improve product descriptions, sizing guides, and quality checks for Fashion and Electronics to reduce returns.
- Optimize shipping operations and reduce delivery delays to improve customer satisfaction and lower return rates.
- Increase marketing efforts and operational improvements in the Central region to improve sales performance.
- Focus on campaigns in high-performing customer segments based on age and gender insights.
- Continue investing in Electronics because it consistently delivers the highest sales and profit performance.

Dashboard and Visualization

dashboards were developed using Power BI to visualize all the analysis and the important insights of the data.

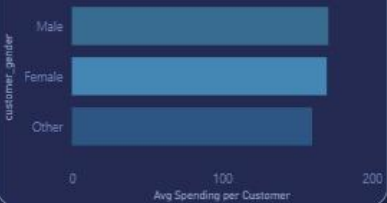


Dashboard and Visualization

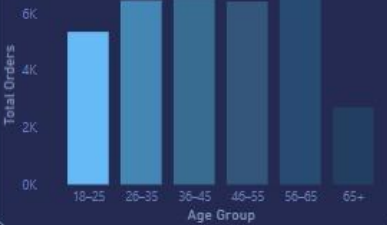
Customer Insights

Demographics, spending behavior, and segment analysis

Avg Spending per Customer by customer gender



Total Orders by Age Group



Avg Spending per Customer by Age Group



customer_gender	Beauty	Electronics	Fashion	Grocery	Home	Sports	Toys
Female	2039	2997	2965	2059	2667	1975	2028
Male	1912	2971	3061	1947	2576	1998	1949
Other	160	231	235	169	211	164	186

Insights:

Male customers dominate in Electronics (38%) and Sports (17%), while female customers lead in Fashion (31%) and Beauty (28%). These insights can support more targeted marketing campaigns and personalized promotional strategies for each customer segment.

EXCLUSIVE OVERVIEW

Overall business performance · 2023-2025

customer gender

All

return

All

payments

All

categories

All

regions

All

5.87M

Total Sales

970.02K

Total Profit

35K

Total Orders

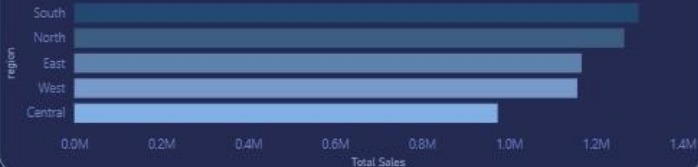
0.05

Avg Discount

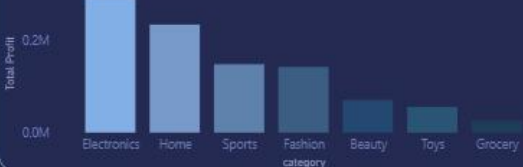
0.06

Return Rate

Total Sales by region



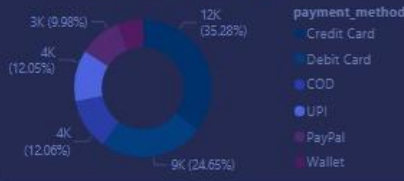
Total Profit by category



Total Sales by Month



Total Orders by payment_method



Dashboard and Visualization

